

Quiz

Questions: 15

Source: organicchemistry.pdf

Date: 11/24/2025

1. Which type of reaction involves the substitution of a hydrogen atom with a nucleophile?

- A) Pericyclic reaction
- B) Addition reaction
- C) Substitution reaction
- D) Elimination reaction

2. What is the primary type of bond found in alkanes and cycloalkanes?

- A) Hydrogen bond
- B) Triple bond
- C) Double bond
- D) Single bond

3. What is the term for the process by which a double bond is formed in an organic molecule through the removal of a hydrogen atom?

- A) Addition reaction
- B) Formation of alkenes and alkynes
- C) Substitution reaction
- D) Elimination reaction

4. What is the functional group that consists of a carbon atom bonded to a nitrogen atom?

- A) Aldehyde
- B) Alkene
- C) Alkane
- D) Amine

5. Which type of bond is found in alkenes and alkynes?

- A) Hydrogen bond
- B) Double bond
- C) Single bond
- D) Triple bond

6. What is the term for the structures of organic molecules that have four single bonds to carbon?

- A) Compounds with four single bonds to C
- B) Cycloalkanes
- C) Alkanes
- D) Alkenes

7. Which type of atom has a high electronegativity and can form bonds with other atoms?

- A) All of the above
- B) Oxygen
- C) Carbon
- D) Nitrogen

8. What is the term for the structures of organic molecules that have double bonds to carbon?

- A) Compounds with double bonds to C
- B) Alkanes
- C) Alkenes
- D) Cycloalkanes

9. What is the functional group that consists of a carbon atom bonded to hydrogen?

- A) Alkene
- B) Alkane
- C) Alkyne
- D) Aldehyde

10. Which type of reaction involves the addition of a nucleophile to a carbon-carbon double bond?

- A) Substitution reaction
- B) Elimination reaction
- C) Pericyclic reaction
- D) Addition reaction

11. Which type of atom has a low electronegativity and can form bonds with other atoms?

- A) All of the above
- B) Nitrogen
- C) Oxygen
- D) Carbon

12. Which type of bond is found in compounds with C=O bonds?

- A) Single bond
- B) Double bond
- C) Hydrogen bond
- D) Triple bond

13. What is the term for the process by which a double bond is formed in an organic molecule?

- A) Formation of alkenes and alkynes
- B) Elimination reaction
- C) Addition reaction
- D) Substitution reaction

14. What is the functional group that consists of a carbon atom bonded to oxygen?

- A) Alkane
- B) Carboxylic acid
- C) Alkene
- D) Aldehyde

15. Which type of atom has unshared electron pairs that can participate in bonding?

- A) Carbon
- B) Nitrogen
- C) Oxygen
- D) All of the above

Answer Key

1. C) Substitution reaction
2. D) Single bond
3. D) Elimination reaction
4. D) Amine
5. D) Triple bond
6. A) Compounds with four single bonds to C
7. A) All of the above
8. A) Compounds with double bonds to C
9. B) Alkane
10. D) Addition reaction
11. A) All of the above
12. B) Double bond
13. A) Formation of alkenes and alkynes
14. B) Carboxylic acid
15. D) All of the above